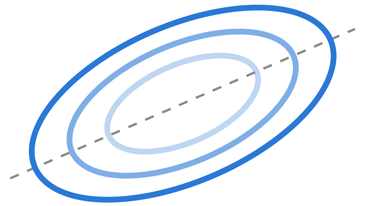
STAT3007 SEMINAR · GROUP G2

Scale or Shape?

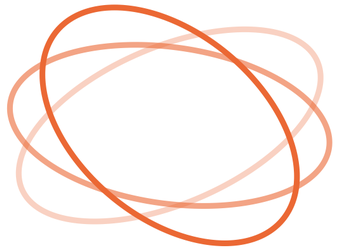
Decomposing the movement of the neural tangent kernel during grokking

Zain Al-Saffi · Jasper Chong · Aaron D'Mello
Zac Kienzle · Samuel Rieger

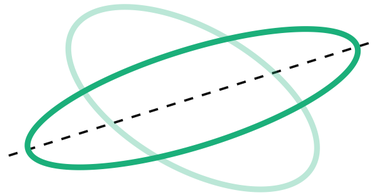
Semester 2, 2026



scale
the kernel grows



shape
the kernel changes shape



alignment
the kernel lines up with the labels

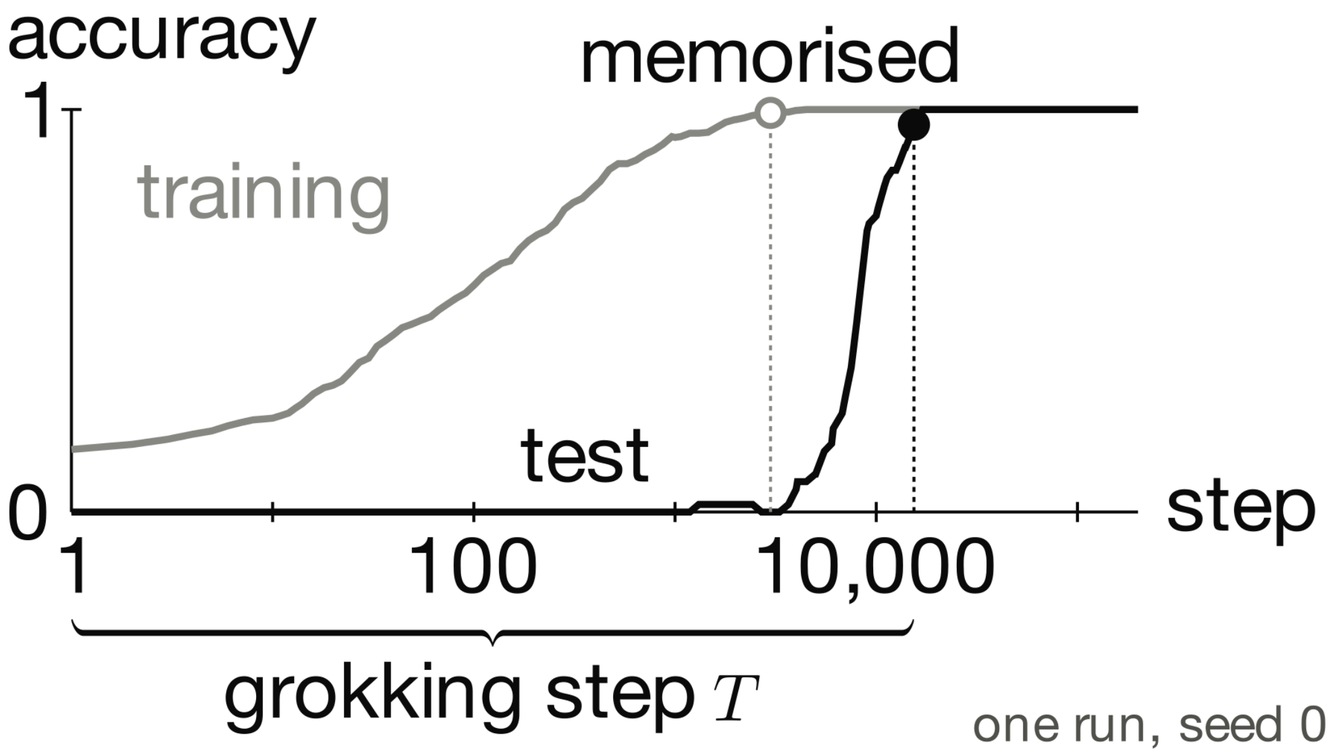
schematic

PART 1 OF 5

Grokking

What does it mean for a network to grok?

A network can fit its training data long before it generalises



MEMORISATION

The first step at which training accuracy reaches 0.99.

GROKKING STEP T

The first step at which test accuracy reaches 0.95.

THE GAP

Over five seeds the median memorisation step is 2,983 and the median grokking step is 23,000.

Grokking was first reported by Power et al. (2022).

The task is modular addition

THE TASK

The input is a pair (a, b) with $a, b \in \{0, \ldots, 22\}$. The target is $(a + b) \bmod 23$, so $(20 + 5) \bmod 23 = 2$.

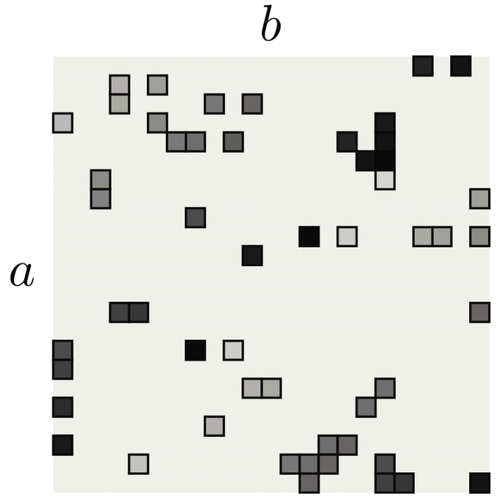
THE SPLIT

Train 476 of the 529 pairs, tests on the other 53.

THE NETWORK

The network has one hidden layer of width N. It is trained on the squared error by full-batch gradient descent.

$$f(x;\theta) = N^{-1/2}\, W^{(2)} \phi\Big(D^{-1/2} W^{(1)} x\Big), \qquad D = 46$$



476 training pairs

53 test pairs

Can the network fill these in?

Modular addition is a common test bed (Gromov 2023; Nanda et al. 2023), and our setup is the baseline of Kumar et al. (2024, App. 8.3, arXiv v3).

PART 2 OF 5

The tangent kernel

What does it mean for a network to learn features, and how could we see it happen?

The tangent kernel shows which inputs the network treats alike

The tangent kernel is a snapshot of the features a network learns with.

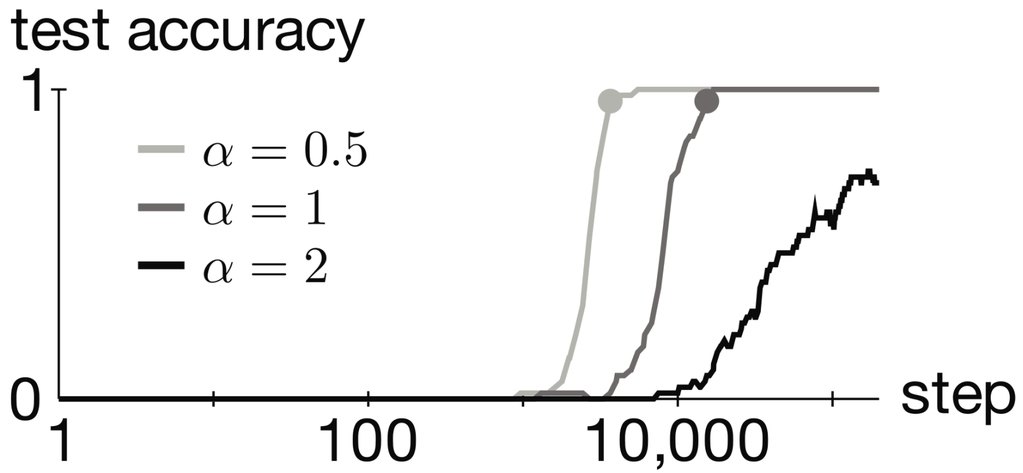
$$K(x, x') = \left\langle \nabla_\theta f(x;\theta),\ \nabla_\theta f(x';\theta) \right\rangle$$

- A gradient step on x moves the output at x' in proportion to $K(x', x)$.
- Near its start the network is a linear model on the features $\nabla_\theta f$ (Chizat et al. 2020).
- If K stays fixed, training ends at kernel regression with K (Jacot et al. 2020).

Grokking may be the move from a fixed kernel to one that learns features

- **Lazy.** The kernel stays near its start, as in kernel regression.
- **Rich.** The kernel changes, and the network learns features.
- Grokking may be the move from lazy to rich (Kumar et al. 2024).
- A larger output scale α keeps the network lazy for longer (Chizat et al. 2020).

$$\tilde{f}(x;\theta) = \alpha\left[f(x;\theta) - f(x;\theta_0)\right], \qquad \eta = \eta_0/\alpha^2$$



A kernel can grow or change shape, and a distance cannot tell which

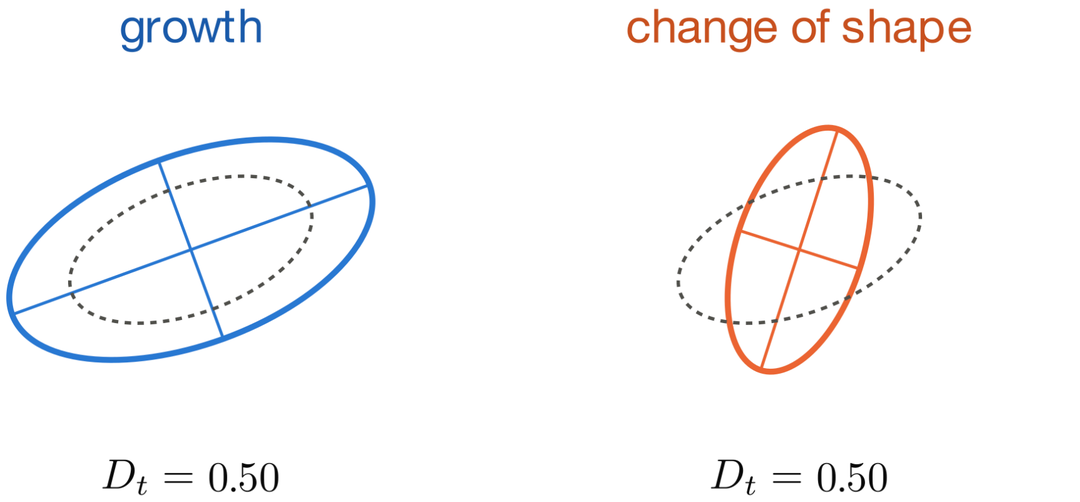
The usual measure of how far the kernel has moved is a distance (Geiger et al. 2020; Mohamadi et al. 2024).

$$D_t = \frac{\|K_t - K_0\|_F}{\|K_0\|_F}$$

It splits exactly into a part for size and a part for shape.

$$D_t^2 = \underbrace{(S_t - 1)^2}_{\text{scale part}} + \underbrace{2\,S_t\,R_t}_{\text{shape part}}$$

> In our baseline, the scale part is 90% of D_t^2 at grokking.



Three numbers say how the kernel moved

SCALE

$$S_t = \frac{\|K_t\|_F}{\|K_0\|_F}$$

The size of the kernel relative to the start, 1 when unchanged.

SHAPE

$$R_t = 1 - \frac{\langle K_t, K_0 \rangle_F}{\|K_t\|_F \, \|K_0\|_F}$$

One minus the cosine between K_t and K_0 when treating them as long vectors. It is 0 when the kernel is only rescaled (Fort et al. 2020).

ALIGNMENT

$$A_t = \frac{\langle CK_tC,\, CYY^\top C \rangle_F}{\|CK_tC\|_F \, \|CYY^\top C\|_F}$$

The cosine between the centred kernel and the label kernel $YY^\top$, which marks pairs with the same answer. It is high when the network treats inputs as alike exactly when they share an answer (Cortes et al. 2024).

Shape and alignment ignore rescaling.

K_t is the summed-logit kernel on the 53 test pairs of a run (Mohamadi et al. 2022).

PART 3 OF 5

What sets the clock?

Which kind of kernel movement sets the grokking time?

Every published clock is scale-like

A **clock** is the quantity that sets when a network groks.

WEIGHT NORM

The network groks once weight decay has shrunk the weight norm to a critical value (Khanh et al. 2026; Khanh 2026).

WEIGHT-DECAY RATE

The grokking time scales with $1/(\eta\lambda)$ (Kim 2026; Pracher et al. 2026).

LAZY TO RICH

Grokking is the move to a kernel that adapts. Its evidence is a distance, which mostly records growth (Kumar et al. 2024).

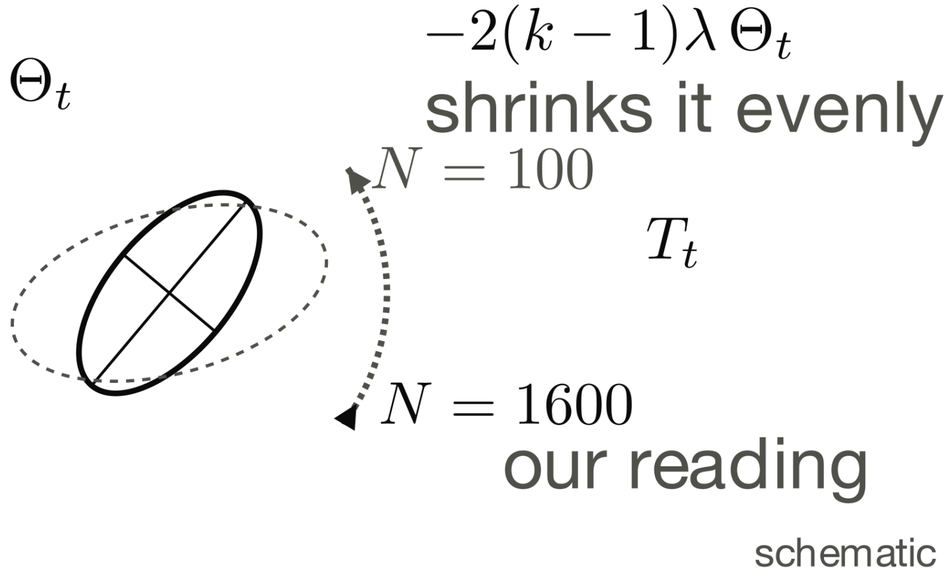
None of them measures the shape of the kernel. We ask which kind of kernel movement sets the time.

In theory, weight decay rescales and width slows the change of shape

$$\frac{dK_t(x,x')}{dt} = \underbrace{-2(k-1)\lambda\, K_t(x,x')}_{\text{scale part}} + \underbrace{T_t(x,x') + T_t(x',x)}_{\text{shape part}}$$

- Weight decay λ enters only the scale part, which shrinks the kernel evenly.
- Only T_t changes the shape, and at the start it is of order $1/N$.
- With $T \propto N^a(\eta\lambda)^b$, a scale clock gives $b = -1$ (Kim 2026). A shape clock gives $a > 0$ and $b \approx 0$ (our prediction).

Eqs. S5 and S6 of Lewkowycz and Gur-Ari (2021, arXiv v2), for gradient flow on a network whose output is homogeneous of degree k in its weights. For our network it is a guide.



Our hypothesis: the shape of the kernel sets the grokking time

Hypothesis. A network groks once its kernel has changed shape enough, so the shape statistic R_t sets the clock.

Test	Shape clock (our hypothesis)	Weight-decay clock	Scale clock
Tier 1. Is a statistic the same at every grokking step?	Yes, R_t	Not stated	Yes, S_t (our prediction)
Tier 2. How does T scale?	Wider networks grok later, and weight decay matters little (our prediction)	$T \propto 1/(\eta\lambda)$, and width does not matter (our prediction)	Not stated
Tier 3. What predicts T early?	R_t	$\eta\lambda$, which a real network does not report	S_t

"Our prediction" marks a prediction that the source does not state, which we derived from it.

We trained 720 networks across laziness, width and weight decay

Setting	Values	What it tests
Output scale α	0.5, 1, 2	How lazy training is
Width N	50 to 1600, six values	How fast the shape can change
Weight decay $\eta\lambda$	0, and seven values from 3×10^{-6} to 3×10^{-3}	The weight-decay clock

Each of the 144 cells has 5 seeds, and each run has 200,000 steps.

A run that has not grokked by then is kept as **censored**, with $T > 200{,}000$. The fits use it through the likelihood (Kalbfleisch and Prentice 2002).

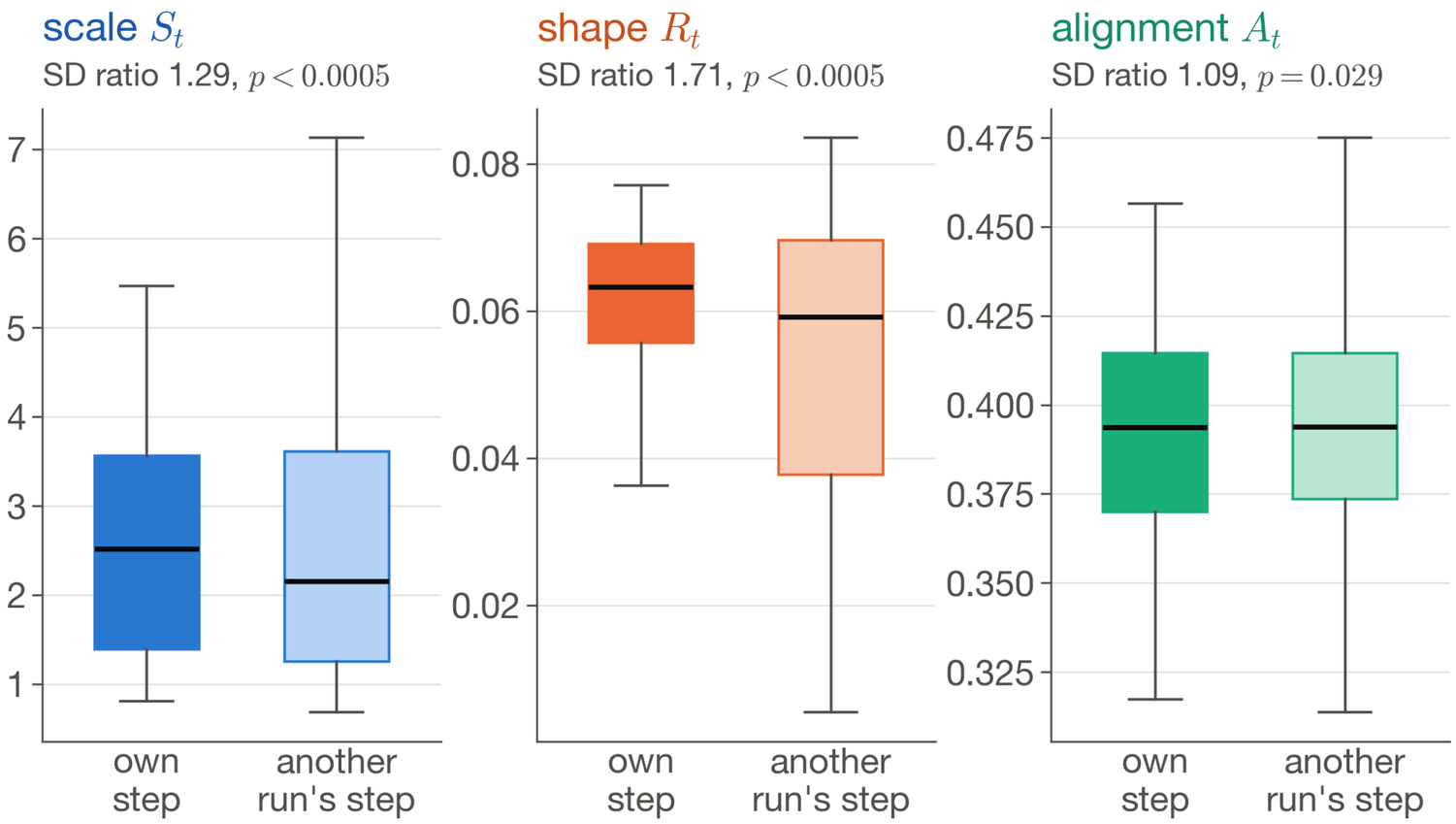
Our baseline grokking steps are close to those of Kumar et al. (2024, fig. 2(c), arXiv v3).

PART 4 OF 5

Results

Which clock does the grokking time follow?

Tier 1: at a fixed width, shape is the statistic most tied to grokking



If grokking has a boiling point, a statistic reads the same in every run at the moment it groks.

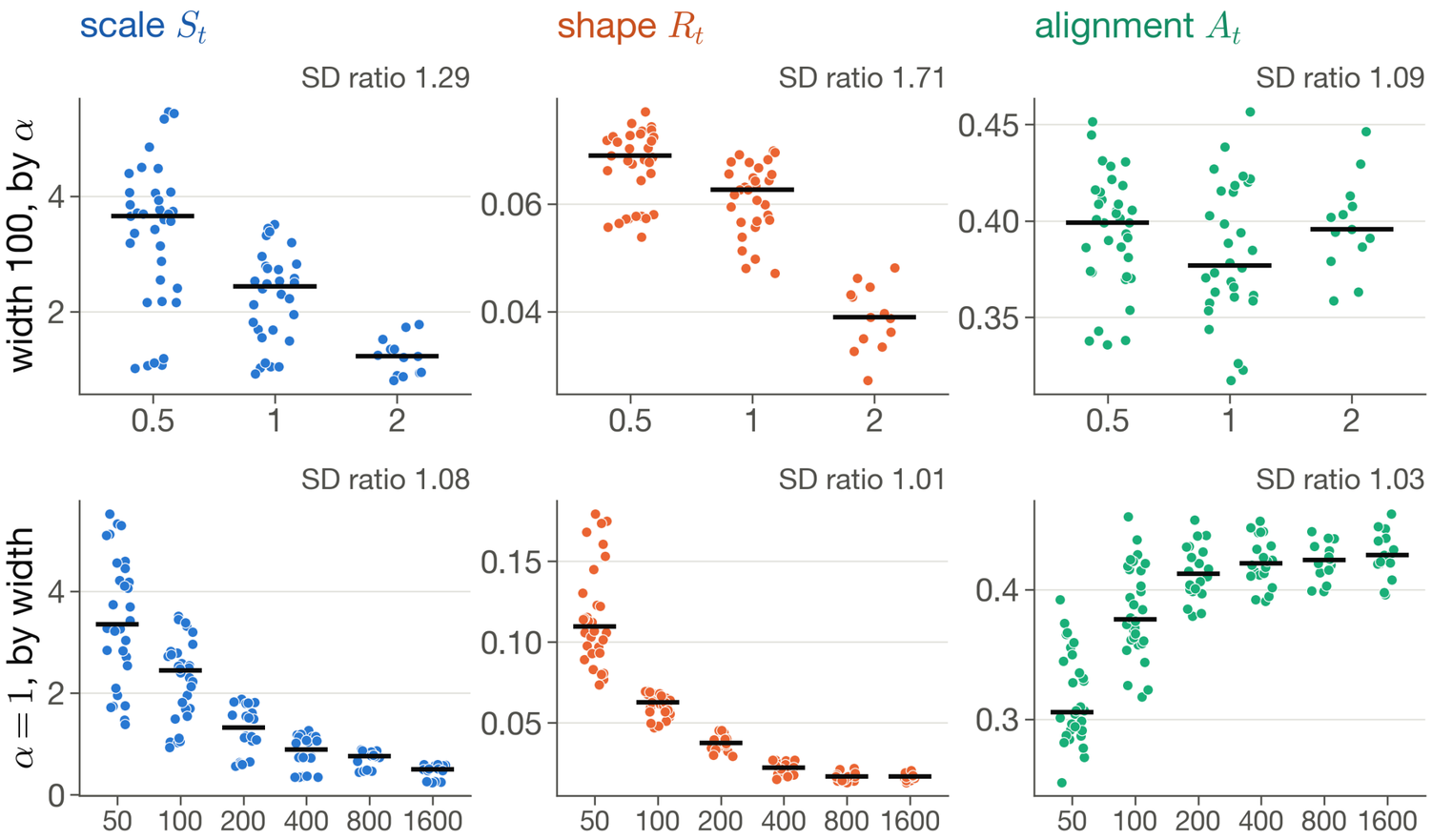
We also read each run at another run's grokking step. The **SD ratio** says how much more the readings spread there.

The 78 runs of width 100 that grok. Boxes hold the middle half of the runs, and whiskers reach 1.5 box heights.

FOR THE CLOCK

Shape bunches up the most at the true grokking step, and then scale. Alignment hardly changes.

No statistic is a fixed threshold, so shape is not the whole clock



- At width 100 (top), shape bunches up at grokking.
- Across widths (bottom), no statistic keeps one value at grokking, and no p is below 0.05.
- Wider networks grok with less change of shape, yet later.

FOR THE CLOCK

Shape tells us the most. Shape alone does not set the grokking time.

Tier 2: the grokking time follows both width and weight decay

$$\log T = c + a \log N + b \log(\eta\lambda) + \gamma \log N \log(\eta\lambda) + \sigma\varepsilon$$

Each clock predicts $T \propto N^{a}(\eta\lambda)^{b}$, a straight line in logarithms. γ lets the settings interact, and its interval **includes 0**.

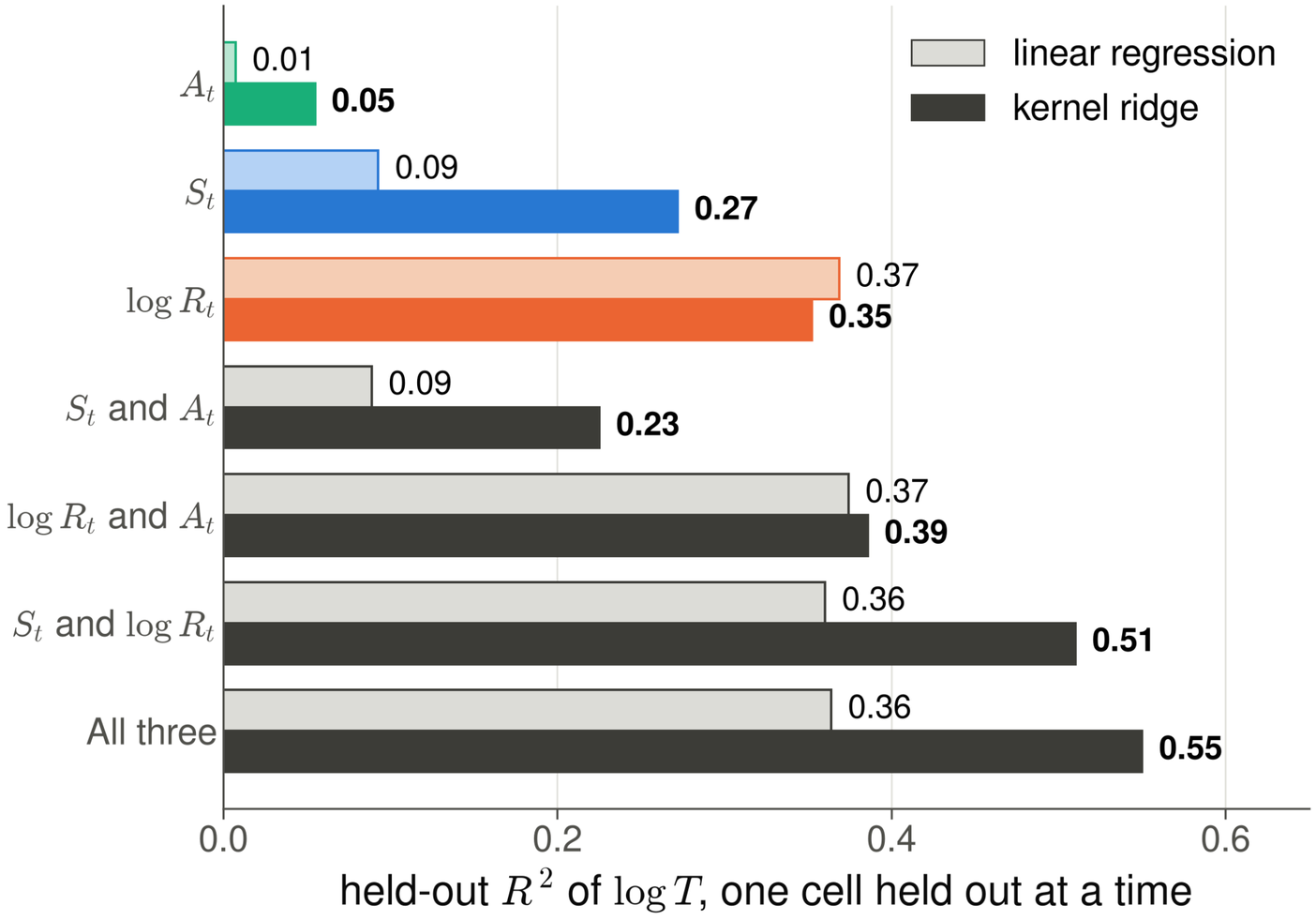
	a **(width)**	b **(weight decay)**	γ	σ
Shape clock predicts	> 0	≈ 0		
Weight-decay clock predicts	≈ 0	-1		
Our fit	$+0.24$	-0.63	-0.16	0.34
95% interval	$+0.08, +0.40$	$-0.92, -0.46$	$-0.37, +0.05$	

FOR THE CLOCK

The fit takes $a > 0$ from the shape clock and $b < 0$ from the weight-decay clock. The clock is mixed.

A censored log-normal fit on the 90 runs at $\alpha = 1$ with $10^{-5} \le \eta\lambda \le 10^{-4}$, where every seed groks.

Tier 3: shape is the best single predictor, and scale adds what shape lacks



We read the statistics at step 2,430, before any run groks, and predict $\log T$ for settings the model has not seen. R^2 is the share of the spread in $\log T$ that the prediction explains.

In a straight line, scale adds nothing to shape. Kernel ridge can bend scale and combine it with shape.

FOR THE CLOCK

Shape alone predicts best. Scale helps once the model can bend it, and then R^2 rises from 0.35 to 0.51. The clock is mixed.

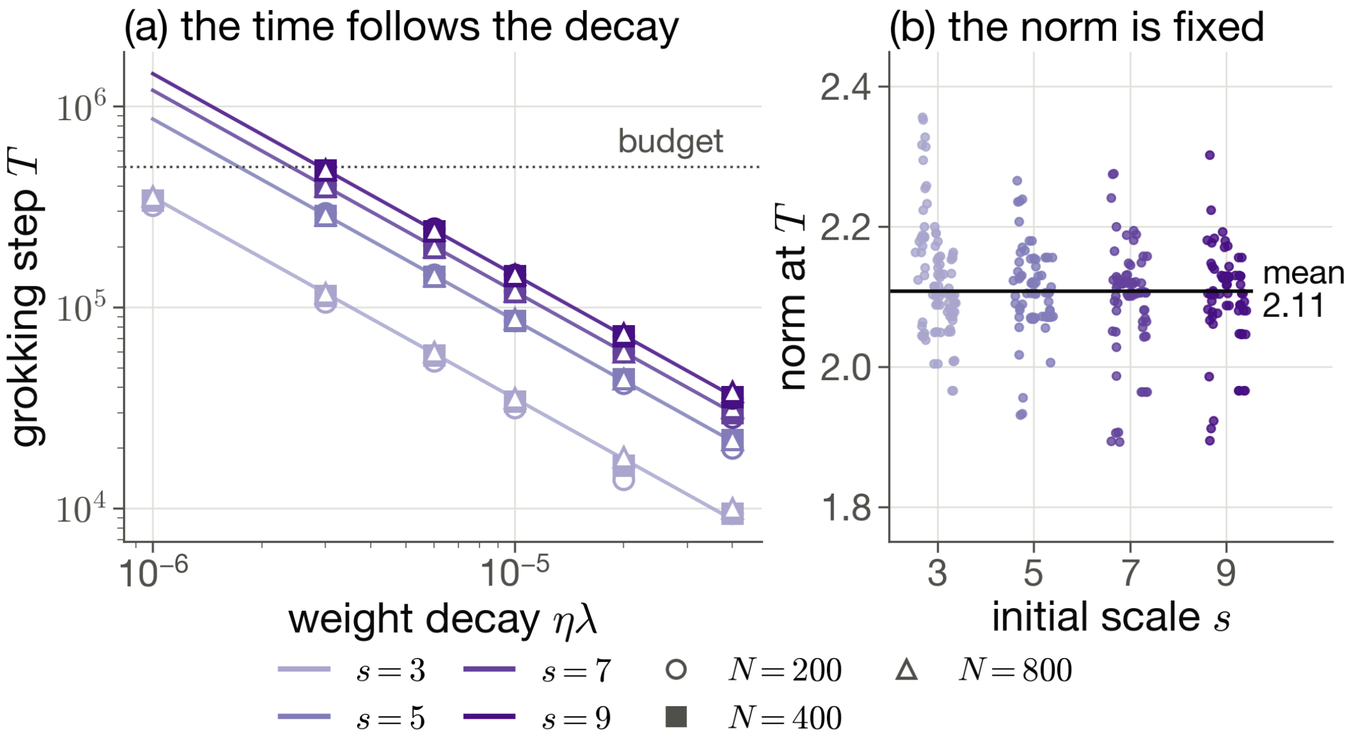
A straight line with scale squared reaches 0.50, and with the products of the statistics 0.54. We use $\log R_t$, since with R_t as it is all three reach only 0.47.

Tier 4: on MNIST, weight decay alone sets the time

We repeated Tiers 1 and 2 on MNIST in the setup of Liu et al. (2023).

Setting	Value
Data	1,000 training images
Network	three layers, rectifier, no biases, squared error
Initial scale s	3, 5, 7, 9
Width N	200, 400, 800 in the fits
Weight decay	seven values, 5 seeds each
Budget and level	500,000 steps, test accuracy 0.80

Large initial weights must shrink to a fixed norm, so $T \propto 1/(\eta\lambda)$ and width drops out.



FOR THE CLOCK

MNIST does not force the kernel to change shape, so only the scale route acts. Modular addition does (Mohamadi et al. 2024), so both act (our reading).

PART 5 OF 5

What we found

So what sets the grokking time?

Grokking runs on a mixed clock, and shape is its best single reading

GROWTH

The usual kernel distance mostly records growth, so split it into scale and shape.

SHAPE

Shape is the most informative statistic. It is not enough alone and not a fixed threshold.

A MIXED CLOCK

No single account holds, ours included. Width, through shape, and weight decay, through scale, both set the time.

On MNIST weight decay alone sets the time. That shape is needed only where a fixed kernel cannot solve the task is our reading (Mohamadi et al. 2024).

Limitations

- One task, one architecture and the NTK parameterisation.
- No seed groks in 70 of the 144 cells, and the exponents depend on which cells are fitted.
- The settings predict better than the kernel statistics.
- MNIST changes the task, the network and how the delay is made at once.

Thank you

Questions?

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